

How Sciences Define Their Objects

Toward a General Architecture of Scientific Explanation

Formal comparative research paper

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Abstract

This paper asks whether mature sciences share a general architecture by which they define and organize primary objects of explanation. A comparative analysis of eighteen natural, social, formal, engineering, and interdisciplinary fields is combined with a review of major accounts of scientific explanation, theory change, scientific objects, natural kinds, mechanisms, levels, experimentation, and disciplinary formation. The comparison does not support a universal object-centered template. Disciplines may be organized around entities, processes, relations, problem classes, methods, or constructed systems; historical sciences need not make precise novel predictions; and mechanism is not the exclusive form of explanation. A weaker but stable architecture does emerge. Mature sciences stabilize inquiry by coordinating seven functions: (1) a delimited domain of phenomena and explanatory targets; (2) units, levels, and scope conditions; (3) ontological commitments concerning relevant entities, properties, relations, and processes; (4) epistemic access through observations, measurements, instruments, records, and interventions; (5) public representations such as classifications, models, diagrams, equations, and simulations; (6) explanatory dependencies expressed as laws, mechanisms, causes, structures, functions, histories, computations, or statistical relations; and (7) success constraints governing evidence, comparison, prediction or retrodiction, intervention, unification, and error correction. These functions are stabilized socially through exemplars, standards, institutions, and trained communities, but community organization is a condition of disciplinary persistence rather than a component of every scientific explanation. The architecture is therefore functional and relational, not a fixed inventory of objects. Applied as a worked example, interaction is structurally compatible with scientific treatment because it can supply phenomena, units, measurements, representations, and candidate dependencies. Structural compatibility alone does not establish theoretical autonomy or justify a new discipline. The answer is qualifiedly affirmative: mature sciences exhibit a general architecture of stabilization, but not a single universal ontology or explanatory form.

Primary contribution: mature sciences do not share one kind of object, but they do converge on a minimal architecture that stabilizes explanatory targets by coordinating ontology, epistemic access, representation, explanatory dependence, and evidential success.

Introduction

Scientific disciplines are often described as if each were defined by a noun: physics studies matter and energy, biology studies life, linguistics studies language, and psychology studies mind and behavior. Such formulations are useful administratively but weak philosophically. They obscure disagreements over what the named object is, which level is primary, how it becomes observable, what forms of explanation count, and why some fields organized around methods or problems nevertheless function as sciences.

The research question is therefore not whether disciplines can be assigned labels. It is whether mature sciences repeatedly solve a common organizational problem: how to select phenomena as explananda, delimit a domain, posit or construct objects, obtain epistemic access, represent dependencies, and evaluate explanations. A genuine general architecture must survive counterexamples from historical, formal, social, engineering, and interdisciplinary sciences. It must also distinguish a scientific object from a scientific discipline. An object may be studied by several disciplines, while one discipline may address several objects.

The paper uses 'scientific object' in a deliberately non-realist-neutral sense: a relatively stable target of systematic inquiry that is individuated through theory and practice and can support intersubjective claims. The definition does not require that every object be directly observable, a natural kind, mind-independent in the strongest metaphysical sense, or unique to one discipline. 'Explanandum' denotes the phenomenon, regularity, event, pattern, capacity, or outcome to be explained. 'Architecture' denotes the smallest recurring set of functions needed to organize sustained explanation, not a historical sequence every science must follow.

Methodologically, the analysis combines structured comparison with philosophical synthesis. The disciplinary survey samples natural, historical, social, formal, applied, and interdisciplinary fields. For each field it identifies the dominant target, unit, ontology, relations, mechanisms or alternative dependencies, evidence, and predictive forms. The philosophical review treats competing accounts as constraints rather than selecting one school in advance. The proposed architecture is retained only where it describes every surveyed field without forcing all sciences into the image of experimental physics.

Problem Statement

A theory of scientific organization can fail in two opposite ways. It can be too strong, requiring a single natural object, universal laws, controlled experiments, or novel predictions and thereby excluding geology, evolutionary biology, parts of astronomy, anthropology, computer science, and design-oriented sciences. Or it can be too weak, listing generic features such as evidence and explanation that distinguish little. The problem is to identify recurring structure without erasing genuine pluralism.

Four distinctions control the analysis:

- Scientific object versus object of a discipline: electrons can be objects of physics, chemistry, materials science, and engineering; no ownership relation follows.
- Ontology versus methodology: what a field claims exists or matters is distinct from how it obtains evidence.
- Explanandum versus unit of analysis: a field may explain population change while measuring organisms, genes, or demographic rates.
- Discipline versus research program: disciplines are durable institutional formations containing multiple programs, whereas programs can cross disciplinary boundaries.

Current Theories of Scientific Organization

Logical structure and explanatory relations

Hempel and Oppenheim treated scientific explanation as subsumption of the explanandum under general laws and initial conditions. The model made explicit the relation between explanans and explanandum, but its symmetry with prediction and dependence on laws proved too restrictive. It remains important because it establishes that explanation requires more than naming a target: a field must articulate relations that make the target expectable or intelligible (Hempel & Oppenheim, 1948).

Subsequent accounts pluralized explanatory dependence. Kitcher emphasized unification through a small set of argument patterns; Salmon emphasized causal relevance and processes; Woodward linked causal explanation to invariance under interventions; and mechanists characterized mechanisms as organized entities and activities productive of regular changes (Kitcher, 1981; Salmon, 1984; Woodward,

2003; Machamer, Darden, & Craver, 2000). These theories disagree about explanation, which is evidence against treating one form - especially mechanism - as architecturally mandatory.

Paradigms, research programs, and disciplinary matrices

Kuhn's mature science is stabilized by shared exemplars, symbolic generalizations, models, and values rather than by a definition of subject matter alone. His later 'disciplinary matrix' is especially relevant: a research community coordinates what problems are legitimate, what solutions look like, and what standards govern puzzle solving. Scientific revolutions can reorganize both ontology and explanatory targets (Kuhn, 1962/1970). Lakatos shifts the unit from paradigms to research programs with a relatively protected hard core and a changing protective belt; progress is judged by theoretical and empirical development over time (Lakatos, 1978). Neither account yields a universal list of objects, but both show that objects are stabilized inside structured practices.

Experimental objects and epistemic things

Hacking argues that scientific entities acquire special warrant through intervention and manipulation, not representation alone. Experimental practice can stabilize entities before theory converges (Hacking, 1983). Rheinberger's 'epistemic things' are materially embodied targets whose relevant properties are not yet fully known; experimental systems generate differences that progressively articulate them (Rheinberger, 1997). Daston and Galison show that scientific objectivity itself has historical forms tied to representational practices and trained judgment (Daston & Galison, 2007). These approaches make instruments and representations constitutive of object stabilization without implying that instruments create every referent.

Natural kinds, classifications, and constructed targets

Essentialist natural kinds cannot organize all sciences. Biological species, psychiatric categories, social classifications, and computational types often have historical, functional, cluster-based, or engineered unity. Boyd's homeostatic property cluster account, Dupré's pluralism, and Hacking's analysis of interactive human kinds show that categories may be stable enough for inquiry without possessing classical essences (Boyd, 1991; Dupré, 1993; Hacking, 1999). Classification is therefore an epistemic achievement to be evaluated, not a universal discovery of fixed essences.

Levels, mechanisms, and perspectival organization

Marr distinguished computational, algorithmic, and implementational levels; explanatory questions change with level. Craver treats levels as component-mechanism relations rather than a single universal hierarchy. Wimsatt emphasizes robustness across independent means of detection and analysis. Giere's perspectivism and Cartwright's attention to capacities caution that models represent selected aspects under conditions, not complete mirrors of the world (Marr, 1982; Wimsatt, 2007; Craver, 2007; Giere, 2006; Cartwright, 1989). A general architecture must therefore include explicit units, levels, and scope conditions.

Communities, boundaries, and institutional emergence

Fleck described thought collectives and styles; Gieryn analyzed boundary work by which communities demarcate science; Abbott showed that disciplines organize through shifting oppositions and problem structures; and Stichweh connected differentiation to specialized communication and institutional reproduction (Fleck, 1935/1979; Gieryn, 1983; Abbott, 2001; Stichweh, 1992). These accounts explain disciplinary persistence and jurisdiction. They do not imply that communities may define arbitrary

objects successfully: empirical resistance, technical performance, and cross-context reliability constrain stabilization.

Comparative Survey of Mature Disciplines

Table 1 records dominant structures rather than exhaustive internal diversity. 'Observer' identifies the epistemic role that selects and records evidence, not necessarily a person external to the system. 'Mechanism/dependency' includes lawful, causal, structural, functional, historical, computational, interpretive, and statistical forms.

Discipline	Primary object / domain	Explanandum	Observer	Unit of analysis	Fundamental entities	Mechanism or dependency	Characteristic questions	Evidence	Typical predictions
Physics	Physical states, fields, spacetime, matter-energy; observable regularities	Events, systems, interactions, distributions	Experimental/theoretical investigator with instruments	System, particle, field, ensemble, spacetime region	Fields, particles, states, observables, symmetries	Dynamical laws, conservation, symmetry, causal/statistical evolution	Why does a state evolve; which invariants constrain it?	Measurement, experiment, astronomical observation, simulation	Quantitative forecasts, parameter estimates, novel effects
Chemistry	Substances, composition, structure, reactions, properties	Transformation and stability of chemical systems	Analytical/synthetic investigator	Molecule, ion, reaction, phase, material sample	Atoms, bonds, orbitals, molecules, phases	Energetic, kinetic, structural, quantum, reaction mechanisms	Why does a substance form/react; how does structure determine property?	Spectroscopy, synthesis, assay, crystallography, computation	Products, rates, equilibria, spectra, properties
Biology	Living systems, heredity, development, function, evolution	Organized persistence, variation, adaptation, lineage change	Observer/experimenter across laboratory and field	Gene, cell, organism, lineage, population, ecosystem by question	Organisms, cells, genes, traits, populations	Mechanisms, selection, inheritance, development, function, history	How is a trait produced, maintained, inherited, or evolved?	Experiment, comparative data, field observation, sequencing, fossils	Outcomes, distributions, phylogenies, functional effects
Ecology	Relations among organisms and environments across scales	Distribution, abundance, interaction, flux, resilience	Field/lab/modeling observer within sampled systems	Individual, population, community, ecosystem, landscape	Organisms, populations, resources, habitats, flows	Competition, predation, mutualism, dispersal, feedback, disturbance	Why are organisms distributed thus; what drives system change?	Surveys, experiments, remote sensing, models, long-term series	Population trajectories, range shifts, community responses
Geology	Earth materials, structures, processes, and history	Present formations and past events/processes	Field, laboratory, and remote observer	Mineral, rock body, stratum, fault, basin, tectonic plate	Minerals, rocks, fluids, landforms, plates	Tectonics, erosion, deposition, metamorphism, geochemical cycles	How did this structure form; what process/history produced it?	Outcrops, cores, dating, geophysics, geochemistry, models	Retrodictions, hazard probabilities, process consequences
Astronomy	Celestial objects, radiation, cosmic structures and evolution	Observed signals and inferred cosmic processes	Earth/space-based observer constrained to remote signals	Photon event, object, orbit, stellar population, cosmological volume	Stars, planets, galaxies, fields, matter, spacetime	Gravity, radiation, fusion, accretion, cosmological dynamics	What is this object; how did it form/evolve; what will be observable?	Imaging, spectra, timing, surveys, gravitational signals, simulation	Transits, positions, spectra, event rates, evolutionary states
Linguistics	Human language as structured capacity, system, use, and change	Patterns in form, meaning, acquisition, variation, processing	Analyst, speaker-judgment participant, corpus/experimental observer	Feature, segment, morpheme, construction, utterance, discourse	Representations, categories, constructions, speakers, communities	Computation, constraint, learning, interaction, historical change	What makes forms possible; how are form and meaning related?	Corpora, elicitation, experiments, typology, historical comparison	Acceptability, parsing, acquisition, change, distribution

Psychology	Behavior and mental processes at individual and group levels	Performance, experience, learning, cognition, affect, development	Experimenter/clinician/participant observer	Trial, response, person, task, group, developmental trajectory	Agents, representations, processes, traits, situations	Learning, memory, perception, decision, motivation, social influence	Why does behavior vary; which processes generate it?	Experiments, tests, observation, reports, physiology, longitudinal data	Response distributions, performance, intervention effects
Sociology	Social relations, institutions, groups, structures, and change	Patterns of coordination, inequality, organization, diffusion	Researcher often embedded in or reflexive toward social systems	Actor, tie, role, group, institution, population, event	Actors, networks, norms, organizations, categories	Interaction, socialization, power, diffusion, selection, institutionalization	How are patterns produced/reproduced; why do institutions change?	Surveys, administrative data, ethnography, experiments, networks	Associations, trajectories, diffusion, policy or institutional effects
Anthropology	Humans in cultural, social, biological, and historical variation	Meaningful practices, lifeways, material and biological variation	Participant-observer, comparative analyst, laboratory investigator	Practice, person, household, community, artifact, population	Persons, symbols, practices, institutions, bodies, material culture	Meaning, exchange, kinship, learning, adaptation, historical formation	How do practices become meaningful and durable; how do populations vary?	Ethnography, archaeology, comparison, biological measures, archives	Contextual expectations, historical reconstructions, pattern contrasts
Economics	Allocation, production, exchange, incentives, institutions, aggregate dynamics	Choices, prices, distributions, growth, strategic outcomes	Analyst using observational, experimental, administrative data	Decision, agent, market, firm, household, economy	Agents, preferences, technologies, goods, institutions, expectations	Optimization, equilibrium, strategic interaction, learning, causal response	How do constraints/incentives generate outcomes; effects of interventions?	Natural/field/lab experiments, panel data, models, national accounts	Comparative statics, causal effects, market and macro trajectories
Computer Science	Computation, information processes, algorithms, artificial systems	Computability, correctness, complexity, behavior of constructed systems	Designer, analyst, verifier, experimenter	Operation, algorithm, program, machine, protocol, data structure	Symbols, states, programs, machines, data, agents	Computation, transformation, communication, learning, control	What can be computed; with what resources; does a system satisfy a specification?	Proof, benchmark, simulation, testing, formal verification	Outputs, resource bounds, correctness, performance, failure modes
Information Science	Creation, representation, organization, retrieval, use, and flow of information	Information behavior and performance of informational systems	Analyst of users, documents, infrastructures, metrics	Document, record, query, user episode, collection, network	Documents, data, metadata, users, systems, institutions	Representation, indexing, retrieval, communication, relevance judgment	How is information organized, found, interpreted, and used?	Logs, retrieval tests, bibliometrics, user studies, ethnography	Retrieval performance, diffusion, use, relevance, citation patterns
Cybernetics	Control and communication in animals, machines, and organizations	Regulation, stability, adaptation, feedback	Observer-controller whose boundary may be inside the system	State variable, loop, controller, system, message	Systems, signals, goals, disturbances, channels	Feedback, feedforward, requisite variety, adaptation	How is stability achieved under disturbance; how does information guide control?	System identification, simulation, experiments, operational traces	Stability, response, convergence, oscillation, controllability
Systems Theory	Organization and behavior of wholes and interdependencies	Cross-domain patterns of organization, emergence, and dynamics	Modeler selecting system boundaries and levels	Component, relation, subsystem, network, whole	Components, relations, boundaries, flows, hierarchies	Interaction, feedback, emergence, constraint, self-organization	Which organizational properties recur; how do parts generate whole behavior?	Models, simulations, comparative cases, network and dynamical data	System regimes, robustness, propagation, emergent patterns
Cognitive Science	Cognition across computational, neural, behavioral, embodied, and social levels	Perception, reasoning, learning, language, action, representation	Interdisciplinary investigator using converging methods	Representation/process, task episode, neural circuit, agent-environment system	Agents, representations, neural systems, bodies, artifacts	Computation, neural dynamics, learning, embodiment, distribution	How is intelligent behavior generated across levels?	Behavioral experiments, neuroscience, modeling, linguistics, AI	Performance, errors, neural/behavioral dynamics, transfer

Human Factors	Human performance and safety in designed sociotechnical systems	Fit among human capabilities, tasks, technology, and environment	Analyst/designer observing operational work	Operator action, task, team, workstation, sociotechnical episode	People, tasks, tools, organizations, hazards	Workload, perception-action, coordination, error, resilience	Why does performance succeed/fail; how do design and organization shape it?	Experiments, field studies, incident records, simulation, task analysis	Performance, workload, errors, reliability, safety outcomes
Human-Computer Interaction	Human use and experience of interactive computational systems	Coupled behavior, usability, experience, appropriation, sociotechnical effects	Researcher/designer; users may be participants and interpreters	Action/event, task episode, interaction sequence, practice, community	People, artifacts, representations, tasks, practices, institutions	Perception-action, cognition, feedback, learning, coordination, meaning-making	How do people and systems mutually shape action and experience?	Experiments, logs, field studies, interviews, prototypes, analytics	Performance, adoption, errors, experience, breakdown, transfer

Recurring Architectural Structures

The survey converges, but not on the candidate inventory in its original form. Object, observer, observation, relationship, mechanism, representation, explanation, prediction, and evidence are not equally fundamental. 'Object' is often plural or processual; 'observer' is sometimes distributed or endogenous; 'mechanism' is one explanatory form among several; and 'prediction' may be replaced by retrodiction, constraint, classification, proof, intervention, or qualitative expectation. The stable commonality is functional.

Component	Function
1. Phenomenal domain and explanatory targets	A field identifies repeatable or historically traceable phenomena, patterns, capacities, events, or problem classes for which descriptions alone are insufficient. The target need not be a single entity.
2. Units, levels, boundaries, and scope	Inquiry specifies what counts as a case, at which scale or level it is analyzed, where the system boundary lies, and under which conditions claims apply.
3. Ontological commitments	The field identifies relevant entities, properties, relations, processes, structures, or constructed types. Commitments may be realist, instrumental, perspectival, or pluralist, but cannot remain entirely implicit.
4. Epistemic access	The field establishes operations by which claims become answerable: observation, measurement, instrumentation, experimentation, intervention, records, comparison, proof, simulation, testimony, or field participation.
5. Public representation	The field stabilizes targets and dependencies through shareable representations: taxonomies, equations, diagrams, maps, models, narratives, code, datasets, specimens, and simulations.

6. Explanatory dependencies	The field states what makes a difference to the target. Forms include laws, causes, mechanisms, constraints, functions, computations, structures, equilibria, selection processes, histories, meanings, and statistical relations.
7. Success and error constraints	Claims face standards that can reveal failure: empirical fit, reproducibility, robustness, intervention, predictive or retrodictive accuracy, proof, unification, coherence, external validity, and comparative explanatory gain.
Stabilizing envelope: research practice	Exemplars, trained judgment, instruments, repositories, standards, institutions, and communities preserve and revise the seven functions. This envelope explains disciplinary durability; it is not itself the explanans in every study.

The architecture can be expressed compactly. A scientific research formation stabilizes a target T when a community can (a) individuate instances of T at specified units and levels, (b) obtain disciplined epistemic access to them, (c) represent their relevant features publicly, (d) propose dependencies that answer contrastive explanatory questions, and (e) expose those claims to shared error-detection and comparison. Ontology, method, and representation are mutually constraining rather than linearly ordered.

Counterexamples

Method-centered fields

Statistics, microscopy, spectroscopy, and parts of computer science appear organized around methods rather than natural objects. They do not refute the architecture because their explanatory targets are problem classes - uncertainty, inference, detection, computation, or measurement - and their objects are often formal or constructed. They do refute the claim that every discipline requires one bounded natural kind.

Historical sciences

Geology, evolutionary biology, cosmology, archaeology, and historical linguistics often explain singular or path-dependent outcomes. Their strongest achievements may be retrodictive consilience, constraint of possible histories, and risky traces expected under one history but not another. They refute prediction symmetry as a universal criterion, not the need for success constraints.

Interpretive and reflexive sciences

Anthropology, sociology, and parts of linguistics investigate agents whose classifications and interpretations can alter the phenomena. Hacking's interactive kinds and reflexive methodology show that the observer-observed relation cannot always be externalized. These fields still establish units, evidence, representations, dependencies, and criticism, but explanation may include reasons, meanings, institutions, and looping effects alongside causes.

Formal and design sciences

Mathematics and theoretical computer science prove consequences within formal systems; design sciences study artificial systems partly by constructing them. Their evidence includes proof and artifact performance, which differ from observation of unaided nature. Simon's sciences of the artificial show that purpose, specification, and synthesis can be legitimate explanatory contexts. These cases require 'epistemic access' and 'success constraints' broad enough to include proof, verification, and constructive demonstration.

Fields without a single paradigm

Psychology, sociology, anthropology, HCI, and cognitive science contain multiple ontologies and explanatory traditions. They remain disciplines because institutional and problem continuity can coexist with theoretical pluralism. Kuhnian consensus is therefore a degree of maturation within programs, not a necessary global state for an entire discipline.

Proposed General Architecture

No accepted philosophy-of-science theory currently specifies how all sciences define their objects. Existing accounts explain different dimensions: logical relations, causal dependence, mechanisms, unification, paradigms, research programs, experimental manipulation, classification, perspectival representation, and community formation. The present proposal is therefore a comparative synthesis, not a discovered consensus theory.

Definitions

Term	Definition
Scientific object	A relatively stable target of systematic inquiry, individuated through coordinated ontological, epistemic, and representational practices, about which intersubjective claims can be made.
Explanatory target	The event, pattern, capacity, regularity, outcome, or contrast for which understanding is sought.
Unit of analysis	The smallest or principal case treated as one instance for a particular inquiry; units may differ across levels.
Epistemic access	The disciplined operations through which evidence relevant to a target is generated, preserved, or evaluated.
Representation	A public structure used to classify, measure, model, simulate, prove, or communicate selected features of a target.
Explanatory dependency	A relation in virtue of which variation, occurrence, organization, or possibility of an explanatory target becomes intelligible.
Success constraint	A standard capable of discriminating stronger from weaker claims and exposing error.

Discipline	A durable social organization of research containing objects, questions, methods, standards, exemplars, and institutions; it need not map one-to-one onto an object.
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Relations among components

1. Target selection is theory-laden but empirically constrained. Phenomena become salient against prior questions and representations, yet recalcitrant observations can force reclassification.
2. Units and boundaries mediate ontology. The same domain can support genes, organisms, populations, or ecosystems as units without one eliminating the others.
3. Epistemic access co-produces object stability. Instruments, records, and interventions determine which properties can be reliably discriminated, while successful manipulation and robustness can increase realist warrant.
4. Representations make comparison possible but are not identical with their targets. Multiple representations may be equivalent for one purpose and incompatible for another.
5. Explanatory dependencies answer contrastive questions under scope conditions. No universal dependency type is privileged across all sciences.
6. Success constraints close the loop by selecting, revising, or rejecting ontologies, measurements, representations, and explanations.
7. Communities stabilize the loop through shared practices, but stabilization remains provisional because standards, instruments, and targets can change.

Necessary and non-necessary components

Candidate	Status	Reason
Object	Retained, generalized	Required as a stable target, but may be a process, relation, problem class, formal structure, or constructed system rather than a natural entity.
Observer	Not a core component	Always practically present, but its location and role vary; better represented through epistemic access and reflexivity.
Observation	Retained, generalized	Observation alone excludes proof, records, simulation, and intervention; replaced by epistemic access.
Relationship	Absorbed	Relations are part of ontology and explanatory dependency.
Mechanism	Not universal	Central in many sciences, absent or secondary in lawful, structural, historical, interpretive, and formal explanation.
Representation	Retained	Required for public, reproducible comparison, though not necessarily pictorial or symbolic in one fixed form.

Explanation	Retained as function	The architecture concerns explanatory organization, but permits plural explanatory forms.
Prediction	Not universal	Replaced by broader success constraints including retrodiction, intervention, proof, and unification.
Evidence	Retained within epistemic access and success constraints	Evidence is indispensable, but its forms and standards differ.

Application to Interaction as a Worked Example

Interaction is examined only after the architecture has been derived. The exercise asks whether it can be organized as a scientific target, not whether it warrants a dedicated discipline or whether any existing ontology is correct.

Architecture component	Possible realization for interaction	Assessment
Phenomenal domain / targets	Reciprocal or sequential changes among coupled agents, artifacts, representations, and environments; coordination, breakdown, adaptation, transfer, and emergent trajectories.	Compatible, but boundaries between interaction, action, communication, control, and activity must be explicit.
Units / levels	Event, action-response pair, state transition, episode, task, practice, relationship, team, institution, or longitudinal trajectory.	Compatible, but no uniquely privileged smallest unit has been established across contexts.
Ontology	Participants or agents, environment, states, actions, signals or representations, opportunities/constraints, histories, and coupling relations.	Existing vocabularies largely suffice; proposed entities remain theory-dependent and should not be treated as proven universals.
Epistemic access	Event logs, observation, experiments, interviews, physiological measures, field records, trace reconstruction, intervention, and simulation.	Strong compatibility; measurement validity and context loss are major risks.
Representation	State-transition models, interaction sequences, graphs, grammars, task models, activity systems, dynamical systems, computational simulations.	Strong plurality; representation equivalence is an open empirical and formal question.
Explanatory dependencies	Feedback, affordances, information constraints, cognition, learning, habit, coordination, social norms, institutional rules, semantics, and technical behavior.	Many candidate mechanisms already belong to adjacent sciences; interaction-level autonomy is unproven.
Success constraints	Cross-context prediction, retrodiction of breakdown, intervention effects, transfer, model comparison, invariance, and explanatory compression beyond component-only accounts.	Compatible; decisive tests require comparison with integrated existing-theory baselines.

The worked example establishes structural compatibility only. Interaction can be individuated as a family of explanatory targets, observed through multiple methods, represented formally, and related to candidate dependencies. However, these conditions are permissive: many legitimate research objects satisfy them while remaining shared across disciplines. To establish theoretical independence, an interaction-centered program would need to show stable interaction-level variables or invariants that produce explanatory and predictive gains not obtainable by integrating cognitive, ecological, informational, social, control, and computational accounts.

A falsifying result would be equally clear: if every robust interaction phenomenon can be explained without residual loss by existing component and relational theories, and if proposed interaction-level constructs add no comparative prediction, intervention guidance, or unification, interaction remains a cross-disciplinary object rather than an autonomous theoretical domain.

Threats to Validity

- Selection bias: eighteen fields cannot represent every science. Medicine, mathematics, statistics, climate science, neuroscience, archaeology, and engineering could shift emphasis, although the counterexample analysis incorporates several of their organizational forms.
- Internal heterogeneity: each table entry compresses contested subfields. The architecture is intended to describe recurring functions, not adjudicate each discipline's ontology.
- Presentist reconstruction: contemporary disciplinary structures may conceal historical alternatives. Kuhnian and historical work partly corrects this, but the survey is not a full genealogy.
- Philosophical underdetermination: realism, instrumentalism, perspectivism, and social constructivism permit different interpretations of stabilization. The framework is designed to remain neutral where the comparative result does not require resolution.
- Functional abstraction: the seven components may appear inevitable because they are broadly formulated. Their value depends on discriminating what is not universal - single objects, mechanisms, external observers, experiments, and prediction - and on specifying relations among the retained functions.
- Application contamination: prior interest in interaction could bias the derived framework. The architecture was therefore formulated and tested against the full disciplinary sample before interaction was evaluated.
- Evidence asymmetry: foundational philosophy sources are not a systematic bibliometric sample. The paper reviews influential competing positions and records disagreement rather than claiming exhaustive consensus.

Discussion

The central result is a qualified convergence. Mature disciplines do not determine their primary objects by discovering a single metaphysically given partition of reality. Nor are objects merely arbitrary products of communities. They are stabilized through a recurrent coupling of explanatory demand, ontological articulation, epistemic access, public representation, and error-sensitive comparison. This coupling can converge on entities such as electrons, processes such as evolution, relations such as ecological interaction, formal structures such as algorithms, or historically constituted practices.

The primary object of a discipline is therefore best understood as a stabilized explanatory target system rather than a noun. 'System' here does not imply systems theory; it denotes the linked set of targets, units, ontological commitments, access procedures, representations, dependencies, and standards that

make sustained inquiry possible. A discipline may host several such systems, and one system may cross disciplinary boundaries.

The distinction between object and discipline resolves apparent anomalies. Cognition is studied by psychology, neuroscience, linguistics, computer science, anthropology, and cognitive science. Information is studied in physics, biology, computer science, communication, and information science. Interaction similarly may be a primary object of particular research programs without being the exclusive object of HCI or the basis of an independent discipline. Disciplinary autonomy requires additional sociological and theoretical conditions: a sustained problem agenda, specialist training, institutions, journals, instruments or datasets, and demonstrable explanatory gains.

The architecture also reframes scientific emergence. New sciences are not born when a label is coined. They emerge when previously dispersed anomalies or questions become a coherent target domain; units and boundaries become reproducible; ontologies support cumulative models; epistemic techniques make relevant differences detectable; explanations outperform competitors; and a community preserves the resulting standards. Instruments and communities can lead this process, but they cannot substitute indefinitely for explanatory success.

No single philosophical account is sufficient. Hempel clarifies the explanandum-explanans relation; Kuhn and Lakatos explain organized continuity and change; Hacking and Rheinberger explain experimental stabilization; natural-kind debates explain classificatory pluralism; mechanism and intervention accounts explain difference-making in many empirical sciences; unification accounts explain cross-domain compression; and sociology of science explains institutional persistence. The proposed architecture is the intersection of these contributions after counterexamples remove non-universal requirements.

Conclusion

Yes, there is a general architecture by which mature sciences define and organize primary objects of explanation, but it is not a universal object ontology or a single theory of explanation. Mature sciences recurrently coordinate seven functions: a phenomenal domain and explanatory targets; units, levels, boundaries, and scope; ontological commitments; epistemic access; public representations; explanatory dependencies; and success and error constraints. Research communities, instruments, exemplars, and institutions stabilize this architecture over time.

The architecture is minimal because removing any retained function prevents cumulative explanatory inquiry: without targets there is no explanandum; without units and scope claims cannot be compared; without ontology relevant differences are unspecified; without epistemic access claims cannot answer to evidence; without representations they cannot circulate or accumulate; without dependencies description does not become explanation; and without success constraints error cannot be discriminated from progress. Conversely, a single natural object, an external observer, experimental manipulation, mechanism, and novel prediction are not universal requirements.

Interaction is structurally compatible with this architecture. That result supports treating interaction as a possible primary object of particular scientific explanations. It does not establish that interaction is theoretically autonomous, that one ontology is correct, or that a dedicated discipline is justified. Those stronger conclusions require comparative explanatory gains that have not been demonstrated by structural analysis alone.

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Appendix A: Evaluation Criteria for a Candidate Scientific Object

Criterion	Question	Failure condition
Target stability	Can researchers identify sufficiently comparable instances or traces?	Cases cannot be individuated except by conclusions the theory is meant to establish.
Unit and scope	Are units, levels, boundaries, and conditions explicit?	Claims shift units or boundaries opportunistically.
Ontological clarity	Are relevant entities, properties, relations, or processes distinguished?	Terms remain metaphorical or collapse into neighboring concepts.
Epistemic access	Can evidence be generated or recovered by disciplined procedures?	No reproducible route connects claims to observations, records, proofs, or interventions.
Representation	Can targets and dependencies be publicly encoded and compared?	Representations are private, unstable, or inseparable from one implementation without justification.
Explanatory dependence	Does the account specify what makes a difference and why?	It merely redescribes or labels observations.
Error sensitivity	Could evidence discriminate the account from alternatives?	No conceivable outcome weakens the claim.
Comparative gain	Does the object-centered organization improve prediction, retrodiction, intervention, unification, or understanding?	Existing integrated theories perform equally with fewer assumptions.